EXPERIMENT-07: Comparison of Two Files for Forensic Investigation Using Compare It

AIM:

To compare two files using Compare It software, identify differences between their contents, and document the findings for forensic investigation.

TOOLS REQUIRED:

- Compare It software
- Windows computer
- Two sample text files
- Screenshot utility or Snipping Tool

Download: Compare It official website.

PROCEDURE:

Step 1: Download and Install

1. Visit the official Compare It website.
2. Download the compatible version of the software.
3. Install the software if an installer is provided, or extract the downloaded package.
4. Launch Compare It.

Step 2: Prepare Sample Files

1. Create two sample text files named file1.txt and file2.txt.
2. Enter similar text in both files.
3. Introduce a few controlled changes in the second file, such as changing words, dates, or sentences.
4. Save both files in an accessible folder.

Step 3: Select Files

1. Open Compare It.
2. Select the first file using the first file selection option.
3. Select the second file using the second file selection option.
4. Click the comparison command to start the analysis.

Step 4: Examine Differences

1. View both files side by side.
2. Observe the color-coded differences.
3. Identify inserted, deleted, or modified text.
4. Record the line numbers and descriptions of important differences.

Step 5: Generate a Report

1. Review the comparison results.
2. Save or export a difference report if supported.
3. Capture screenshots of the file selection and comparison results.
4. Record the findings in the lab observation table.

Step 6: Verify File Integrity

1. Preserve the original files.
2. Calculate SHA-256 hashes of both files if required.
3. Record the hashes before and after examination when integrity verification is needed.
4. Remember that different hashes indicate different file contents, but do not independently establish when or why the files changed.

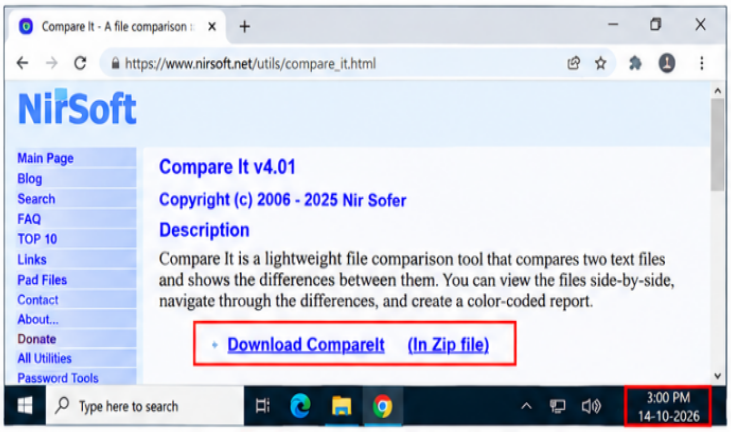


Figure 1: Downloading Compare It from NirSoft Website

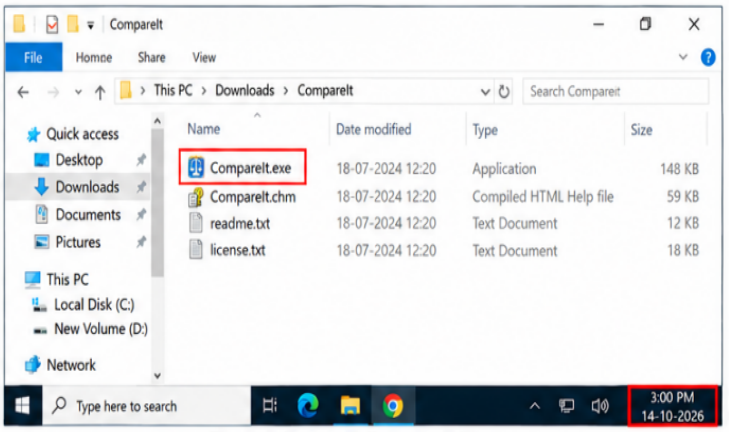


Figure 2: Extracted Compare It Files

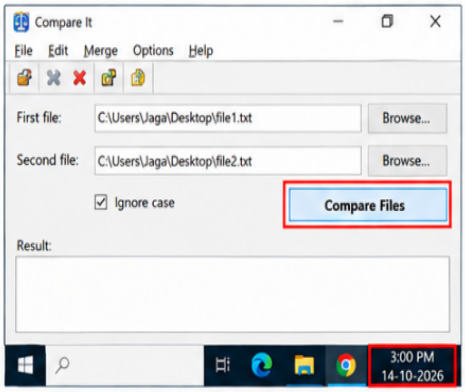


Figure 3: Selecting Two Files in Compare It

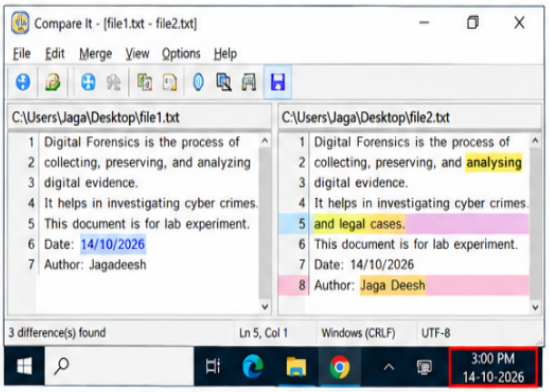


Figure 4: Comparing Files and Highlighting Differences

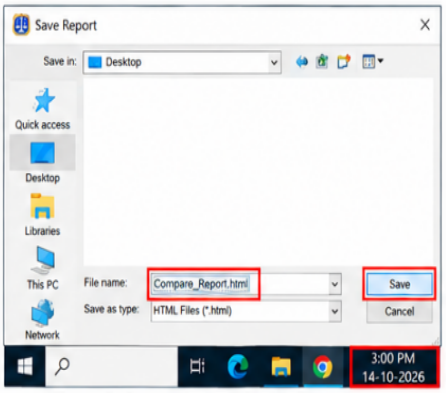


Figure 5: Generating and Saving Report

RESULT:

1. The software successfully identified and highlighted differences between the two files.
2. The side-by-side view with colored markings made it easy to locate changes.
3. The generated reports help document file differences, useful for forensic analysis and evidence presentation.